

# DecodeLabs Data Analytics Internship - Project 2

## Exploratory Data Analysis (EDA) & Business Insights Report

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**Domain:** Data Analytics  
**Status:** Completed & Validated

### 1. Project Objective

The objective of this project was to perform Exploratory Data Analysis (EDA) on the standardized "Gold Standard" online store orders dataset. By implementing descriptive statistics and categorical aggregations, we uncover underlying patterns in performance metrics, measure transaction distributions, identify sales outliers, and provide actionable evidence to optimize operational decisions.

### 2. Statistical Center of Gravity (Five-Number Summary)

The baseline metric audit of our 1,200 unique records reveals a right-skewed revenue distribution structure:

- **The Revenue Floor (Minimum):** \$11.39
- **Lower Quartile spending (25%):** \$410.52
- **The Center Point (50% Median):** \$823.62
- **Upper Quartile spending (75%):** \$1,578.48
- **The Revenue Ceiling (Maximum Anomaly):** \$3,456.40

**Key Insight:** The overall **Mean Total Price (\$1,053.97)** sits significantly higher than the **Median (\$823.62)**. This mathematical gap proves our data is right-skewed. High-spending enterprise or bulk consumer checkouts pulling near the \$3,456.40 ceiling are pulling the standard averages upward, while 50% of our daily transactions naturally sit below \$823.62.

### 3. Core Operational Data Aggregations

#### A. Product Revenue Generation Matrix

| Product Class | Units Sold (Quantity) | Total Revenue Generated | Performance Tier |
|---------------|-----------------------|-------------------------|------------------|
| Chair         | 562                   | \$195,620.11            | High Performer   |
| Printer       | 542                   | \$195,612.61            | High Performer   |
| Laptop        | 535                   | \$192,126.56            | High Performer   |
| Tablet        | 497                   | \$186,568.95            | Mid-Tier         |
| Monitor       | 480                   | \$175,651.41            | Mid-Tier         |

| Product Class | Units Sold (Quantity) | Total Revenue Generated | Performance Tier |
|---------------|-----------------------|-------------------------|------------------|
| Desk          | 508                   | \$167,459.93            | Lower Performer  |
| Phone         | 411                   | \$151,722.39            | Lower Performer  |

- **Observation: Chairs** and **Printers** represent our core asset portfolio anchors, yielding over \$195,600 each. A deeper examination shows that Chairs edge out Printers by a microscopic margin of just \$7.50. While **Phones** remain a common customer item, they generated our lowest total category revenue share (\$151,722.39) due to tighter purchasing volume blocks.

## B. Traffic Channel Customer Acquisition

| Marketing & Referral Source | Acquired Transaction Volume |
|-----------------------------|-----------------------------|
| Instagram                   | 259                         |
| Email                       | 250                         |
| Google                      | 241                         |
| Facebook                    | 228                         |
| Direct Referral             | 222                         |

- **Observation:** Acquisition volume is highly competitive across channels. However, **Instagram** leads customer engagement and conversions with 259 unique transactions, indicating high return-on-ad-spend (ROAS) on visual media formats.

## C. Settlement Infrastructure & Transaction Scaling

| Settlement Channel (Payment Method) | Average Basket Value (AOV) |
|-------------------------------------|----------------------------|
| Credit Card                         | \$1,127.55                 |
| Gift Card                           | \$1,070.97                 |
| Cash                                | \$1,056.04                 |
| Online                              | \$1,017.22                 |
| Debit Card                          | \$1,001.56                 |

- **Observation:** While customers using Debit Cards spend smaller amounts per order (\$1,001.56 average), **Credit Card users represent our highest value shoppers**, pushing an Average Basket Value of **\$1,127.55**.

## 4. Execution Framework

The extraction script executes calculations sequentially via **pandas**. To review or reproduce the terminal output, run:

```
python project2_eda.py
```